

Chemistry By Avinash Singh

* कौश (Shell) $\swarrow$ उपकौश (Subshell)
 K, L, M, N

	s	p	d	f
	↓	↓	↓	↓
$e^- \rightarrow$	2	6	10	14

Max
(अधिकतम)

* इलेक्ट्रॉनिक विन्यास (Electronic Configuration) *

* इलेक्ट्रॉनों को भरना (To fill the e^-)

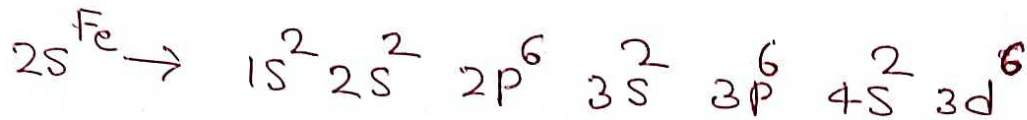
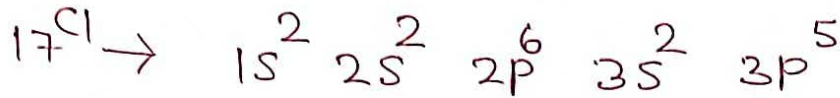
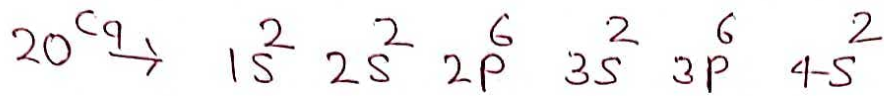
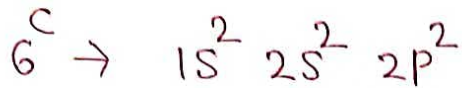
1s 2s 2p 3s 3p 4s 3d 4p

उदाहरण — $_{10}\text{Ne} \rightarrow 1s^2 2s^2 2p^6$

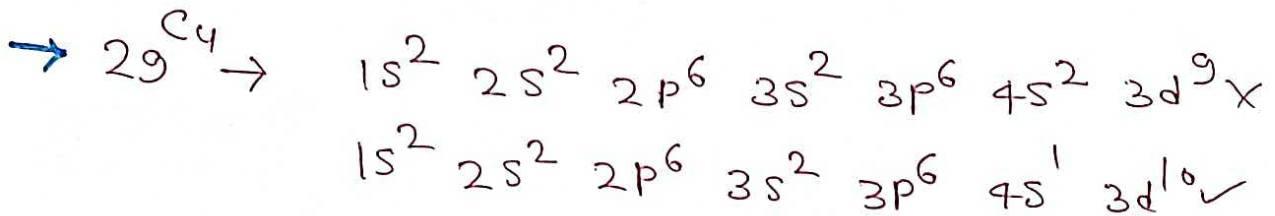
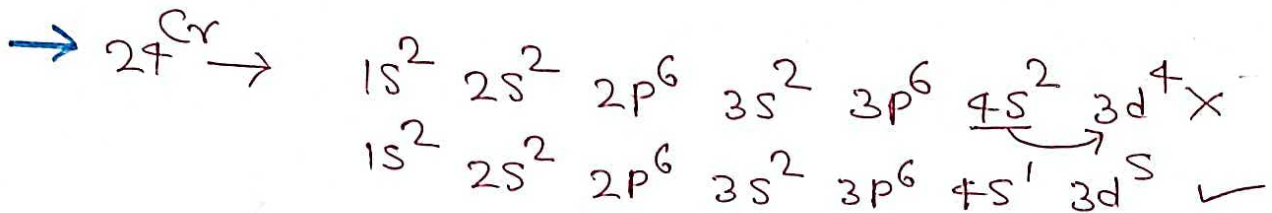
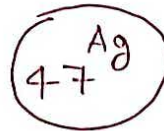
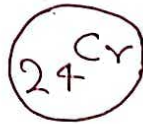
$\text{Ar} \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6$

→ ऑफबाउ नियम के अनुसार — उपकौश की बढ़ती ऊर्जा के क्रम में भरे जाते हैं

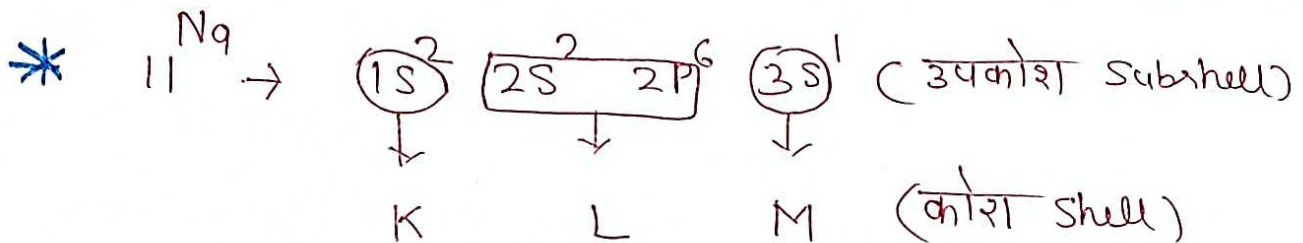
According to Aufbau principle e^- are filled in Subshells in increase order of energy



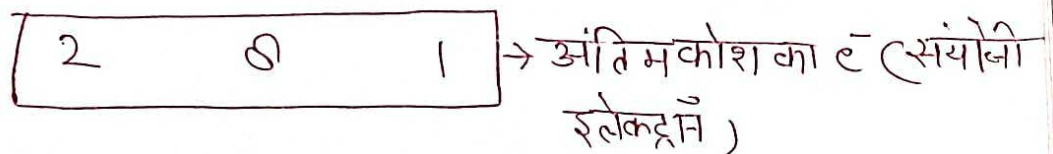
विशेष अपवाद
Note Exception



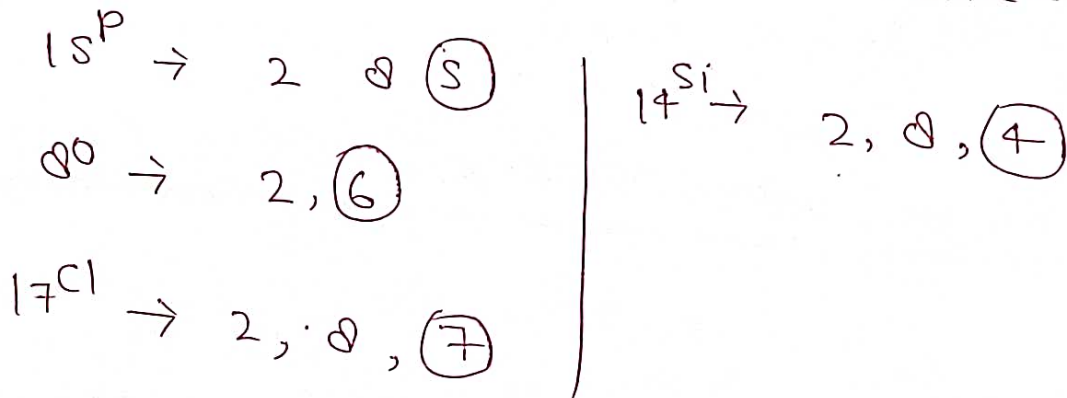
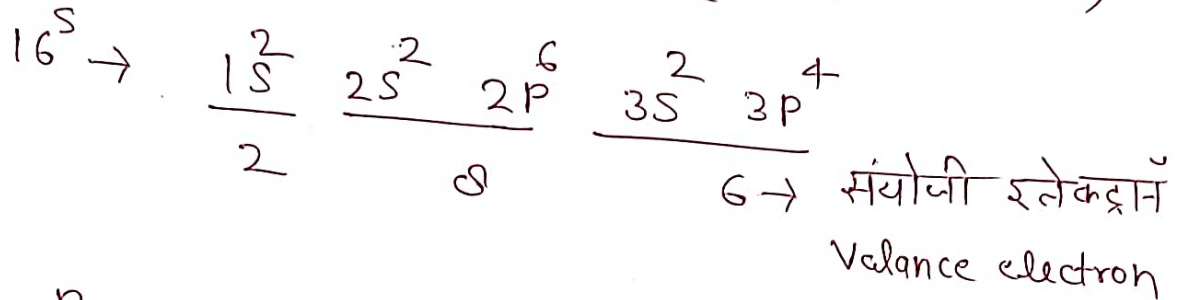
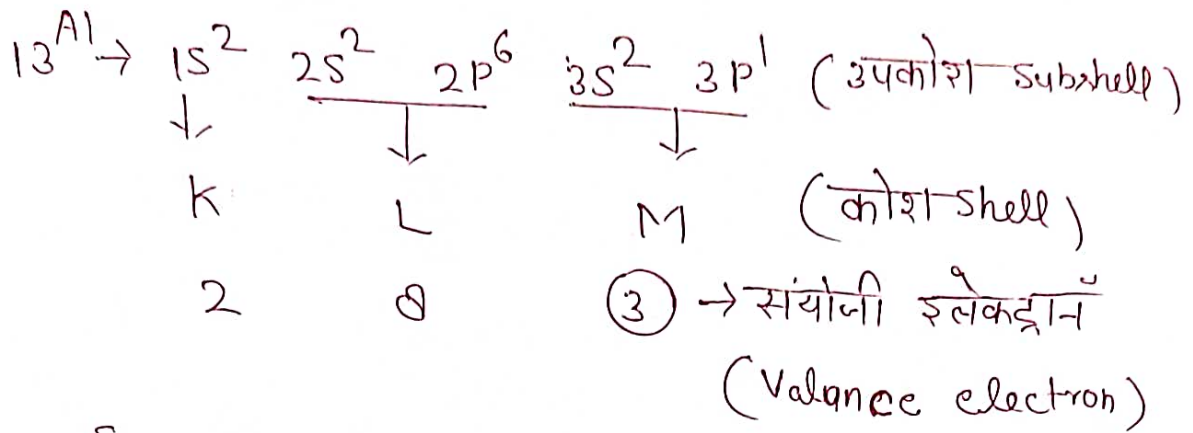
(स्थायी)



(कोश Shell)

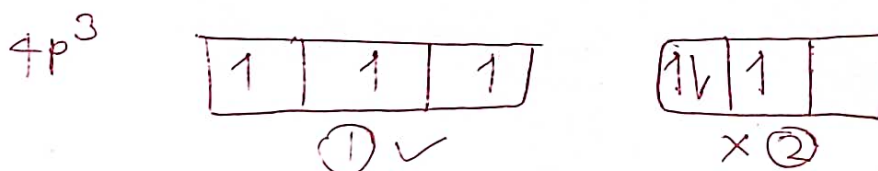
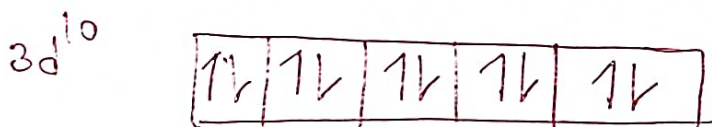
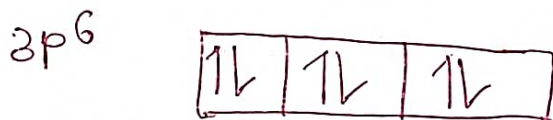


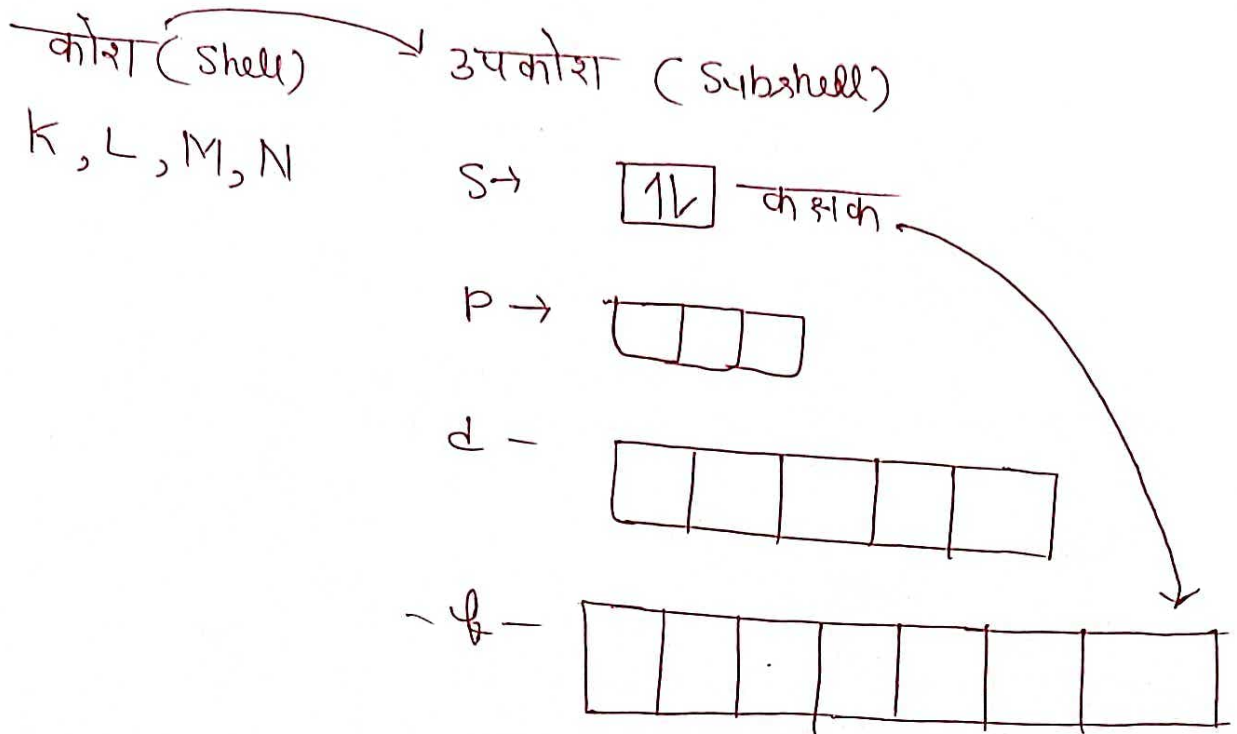
Last Shell e^- (Valance electron)



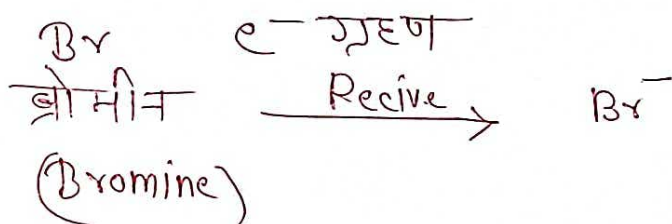
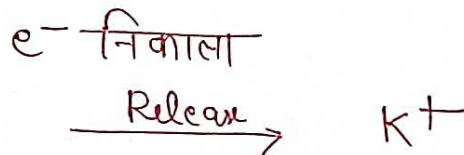
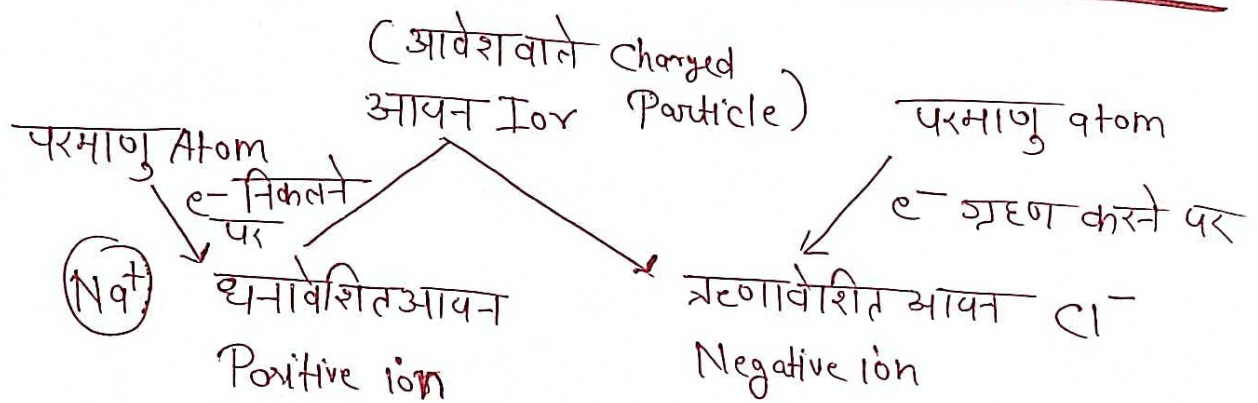
हुण्ड का नियम (Hund's law) -

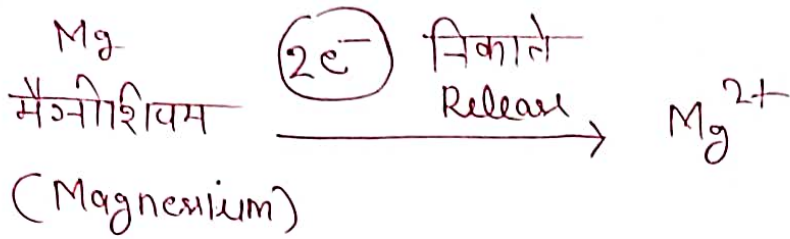
कक्षको में इलेक्ट्रॉन पहले युग्मित भरे जाते हैं फिर युग्मित किये जाते हैं (e^- are filled unpaired then paired in orbitals)



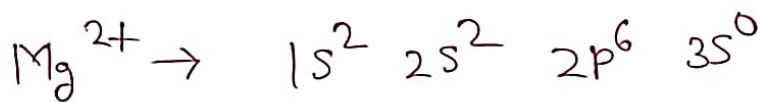
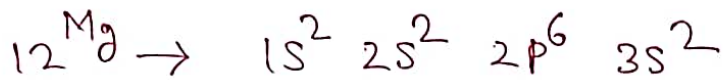


$\rightarrow$ आयनों का इलेक्ट्रॉनिक विन्यास Electronic Configuration of ion

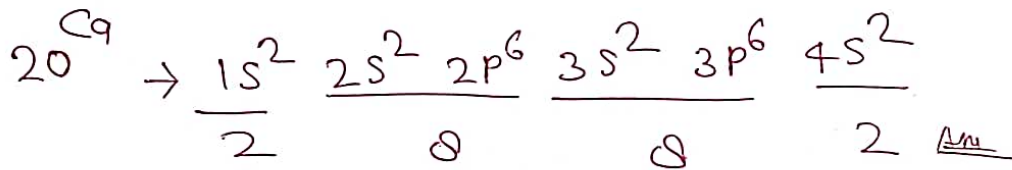




Mg^{2+} का इलेक्ट्रॉनिक विन्यास (Electronic Configuration)



Question Ca (Z=20) का इलेक्ट्रॉनिक विन्यास है



Question निम्न में से ऊनी स्तर का कौन सा क्रम सही है



Question निम्न में से कौन $2p^4$ के लिए सही है

